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SUBJECT: Project Definition for BevoTag, an Open Asset-Tracking Tag for the UT Austin Campus

INTRODUCTION

This Project Definition states the design problem that we, the BevoTag team, will solve in Senior Design and explains how we have organized ourselves to solve it. BevoTag is a small tag that a student or staff member attaches to a laptop, bag, or piece of university equipment to locate it anywhere on the University of Texas at Austin campus. Global Positioning System (GPS) signals do not penetrate well indoors [1, Sec. I], and consumer tags such as the Apple AirTag update an item's location only when a nearby phone from the same manufacturer detects them [2]. BevoTag is an open tag because it needs no manufacturer's phone network: it estimates the item's position from campus Wi-Fi and its own motion sensor and reports that position to its owner on a campus map. We will build a tag, firmware, a building survey, and an application that together place an item in the correct room in more than 90% of trials. We have chosen our technical approach and will next build a data logger for our first Wi-Fi and motion measurements.

Our design problem is to track an item continuously, indoors and outdoors, without new infrastructure. Wi-Fi measurements alone can mislocate a device by several meters because signals fluctuate as the environment changes [1], [3, Table 4], and a low-cost motion sensor alone drifts by more than 150 m within 60 s [4, Sec. 6.3], so we combine the two and let each correct the other. Beyond room identification, the system must keep its median position error under 5 m, update an item's location at least every 10 s while the item moves, and run at least 24 hours on one battery charge. Radio, data-security, and battery-safety standards and university network policy constrain the design, and we commit to recording no one's network traffic, to encrypting and limiting location data, and to using the campus network only as an ordinary client. We will

test the system in surveyed areas beginning with the Engineering Education and Research Center (EER). Esteban Martinez and Li Shi will build the tag hardware and firmware, Andrew Yin and Abhirit Das will build the localization software and the user application, and Ethan Chua will coordinate the work as Project Lead.

PROBLEM STATEMENT

BevoTag must locate a tagged item to within a few meters across indoor and outdoor areas of campus, using only existing campus Wi-Fi and its own motion sensor. Buildings block and reflect satellite signals [5], and low-cost motion sensors drift [4], so no single source of position is reliable on its own. Fusing Wi-Fi fingerprints, the signal patterns recorded at known locations, with motion sensing is the most promising approach, and a successful design must meet measurable targets for room accuracy, update rate, and battery life.

Design Problem Description

A successful BevoTag design must estimate an item's location indoors and outdoors without infrastructure we cannot control, and it must show that location to the item's owner on a map. Existing trackers fail this need because satellite signals do not reach indoors and crowdsourced tags depend on passing phones. BevoTag addresses the need by fusing Wi-Fi fingerprints with motion sensing, an approach that brings three main challenges: computing the position within the tag's limited power, capturing CSI from radio signals that change from moment to moment, and correcting sensor drift. Because the tag reports location, the design must also satisfy radio, data-security, and battery-safety standards and protect the privacy of bystanders and owners.

The Need for On-Campus Item Tracking

Students, faculty, and staff need a way to find misplaced items anywhere on the main UT Austin campus, known as the Forty Acres. Laptops, bags, and shared university equipment move between classrooms, labs, libraries, and outdoor plazas many times a day, and an item left behind can be anywhere among dozens of multistory buildings. A useful tracker must therefore work indoors, on any floor, and in the narrow outdoor spaces between tall buildings. Our goal is tracking that needs neither new infrastructure, such as beacons installed in every building, nor a passerby who owns a particular brand of smartphone.

Why Existing Solutions Fall Short

Each of the three existing approaches to tracking fails on campus. Satellite positioning cannot reach indoors, crowdsourced Bluetooth tags stop updating when no compatible phone passes by, and Wi-Fi positioning that relies on signal strength alone often errs by enough to place an item in the wrong room.

Satellite positioning fails indoors and degrades between tall buildings. Global Navigation Satellite Systems (GNSS), the family of satellite positioning systems that includes GPS, compute a receiver's position from faint signals broadcast by satellites in medium Earth orbit; GPS satellites, for example, orbit about 20,200 km above the Earth [6]. These signals do not penetrate well indoors [1, Sec. I]. Between tall buildings, signal blockage and multipath, the arrival of signals only after they reflect off nearby walls, degrade GNSS so much that one 2009 experiment with a precise receiver lacking inertial aiding measured root-mean-square position errors greater than 9 m [5, Sec. 1.1.1].

Crowdsourced tags locate an item only when a compatible phone happens to pass nearby. The Apple AirTag and the Samsung Galaxy SmartTag2 broadcast a short-range Bluetooth signal, and nearby phones from the same manufacturer detect that signal and report its location [2], [7]. In an empty lab at night no such phone passes by, so the tag's location stops updating until one does.

Wi-Fi positioning that relies only on signal strength is too coarse for reliable room-level tracking. These systems estimate position from the received signal strength indicator (RSSI), a measure of how strong each nearby Wi-Fi signal is when it reaches the device. Signal strength fluctuates as people move and the environment changes [1, Sec. III-B], so RSSI fingerprinting typically locates a device only to within 1 to 5 m [3, Table 4], which can place it in the wrong room.

Theory of Operation

BevoTag combines two sources of position information whose errors differ: campus Wi-Fi measurements, which are noisy but do not drift, and a motion sensor, which is smooth over short periods but drifts. The final design may compute this estimate on the tag or on a server; we will decide after measuring the prototype hardware, as the Design Challenges explain.

The system locates the tag by matching its Wi-Fi measurements against a survey of the surrounding building. In addition to RSSI, some Wi-Fi chips report channel state information (CSI), the frequency response of each of dozens of narrow sub-carriers [8], which shows how much the path weakens and delays the signal on each sub-carrier. Walls, furniture, and room geometry shape these values, so CSI acts as a detailed fingerprint of a location [9]. We will first survey a building by recording Wi-Fi measurements at many points whose positions we know. The system can then estimate the tag's position by finding the survey points whose measurements best match the tag's current measurements.

A motion sensor tracks the tag between Wi-Fi fixes but drifts when it runs alone. An inertial measurement unit (IMU), similar to the sensor that rotates a smartphone's screen, measures acceleration along three axes and rotation about them. Adding up these measurements from a known starting point, a technique called dead reckoning, tracks the tag's movement many times per second. Small sensor errors accumulate, however, so the estimate drifts farther from the truth the longer it runs: a simple low-cost inertial system can accumulate more than 150 m of average position error within 60 s [4, Sec. 6.3].

A fusion algorithm combines the two sources so that the result neither jumps like Wi-Fi alone nor drifts like the IMU alone. A standard candidate is the extended Kalman filter [10], which uses the IMU to predict how the tag has moved and uses each new Wi-Fi fix to correct the accumulated drift, weighting each source by how reliable it currently is. Outdoors, where satellite signals are available, the same algorithm can also accept a GNSS position as an optional third source.

Design Functionality and Scope

BevoTag is a system of a tag, a location server, and a user application that together turn Wi-Fi and motion measurements into a position on a campus map. The project covers a working prototype that we will test in surveyed areas, starting with EER. It excludes cellular connectivity, a campus-wide survey, mass-production tooling, and the tracking of people.

Any BevoTag design must measure, transmit, estimate, and display, as Figure 1 on the next page shows. The tag measures nearby Wi-Fi signals and its own motion (step 1) and sends

these measurements over the campus Wi-Fi network to the location server (step 2). The server combines them with a previously recorded survey of the building to estimate the tag's position, or it receives an estimate that the tag computed itself (step 3). The user application then displays that position on a campus map (step 4). To deliver this behavior, any BevoTag design must perform four functions:

- **Wi-Fi measurement:** record the RSSI and, where the hardware supports it, the CSI of signals from nearby campus access points in the 2.4 GHz band.
- **Motion tracking:** sample the IMU's acceleration and rotation fast enough to follow a walking person carrying the tag.
- **Position estimation:** fuse the motion data with Wi-Fi position fixes, correcting the drift that accumulates in the motion estimate.
- **Reporting:** send the latest position and device status to an application that the owner can view.

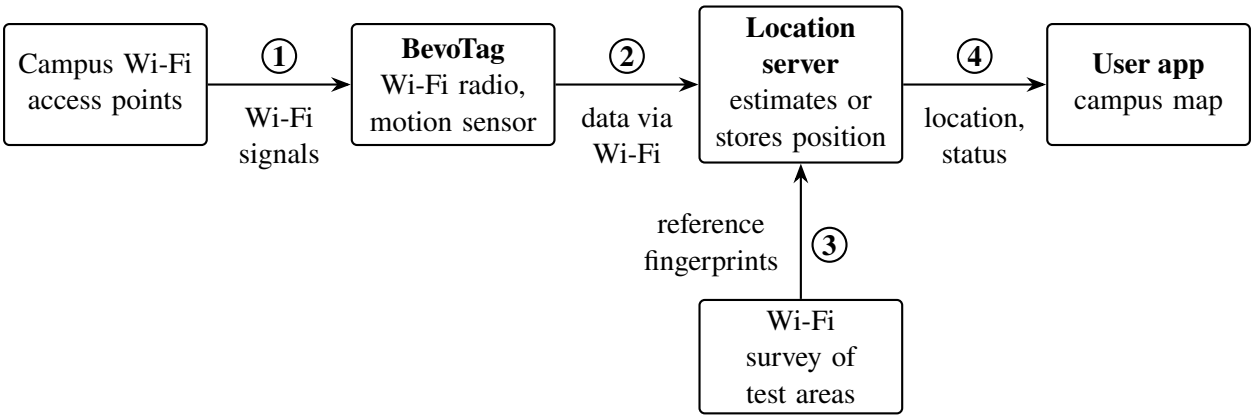


Figure 1. How BevoTag Locates a Tagged Item

Within that scope, the prototype will use available parts and start with one building. The tag will combine a Wi-Fi chip that can report CSI, such as the ESP32-S3 [8], with a low-noise IMU on a development board or a custom circuit board. Its firmware will read the sensors and either process the measurements or send them to the server. The survey will begin with EER and its adjacent outdoor plazas, and the application will display the tag's position on a campus map.

Four related capabilities fall outside the project's scope. The tag will not use a cellular Internet of

Things connection, such as Long-Term Evolution for Machines (LTE-M) or Narrowband Internet of Things (NB-IoT), because these require a paid data plan and a second radio that would raise cost and power consumption, and campus Wi-Fi serves the target areas; the survey will confirm that coverage. We will not survey every building on campus, because a survey requires walking each floor and recording measurements at many points. We will instead test in a few defined benchmark areas and document the survey method so that others can add buildings later. We will 3D-print or buy the prototype enclosure rather than design it for injection molding, since the goal is a working proof of concept rather than a manufacturable product. Finally, BevoTag locates objects, not people, and the design excludes features that would follow a person, such as continuous covert tracking.

Design Challenges

Three challenges will most strongly shape the design: deciding where to compute the position, capturing CSI from radio signals that never repeat exactly, and correcting motion-sensor drift quickly enough. Each challenge requires a measurement on prototype hardware before we can finalize the design.

The first challenge is deciding whether to compute the position on the tag or on the server. Processing CSI and running a fusion filter on a small microcontroller strain its memory and battery. Sending raw measurements to the server moves that work off the tag but keeps the radio on longer, which also drains the battery. We will settle this trade-off by measuring CSI capture rates, IMU data rates, and power draw on the prototype.

The second challenge is capturing reliable CSI from signals that vary from moment to moment. A receiver derives CSI from the training fields of frames that use orthogonal frequency-division multiplexing (OFDM) [8], [11], so the tag must capture it from OFDM frames that nearby access points send, not from every frame it hears. The measurements recorded at a given spot also fluctuate with noise and with changes in the environment, such as people walking through hallways [1, Sec. III-B]. We will therefore record survey measurements at several times of day and test the matching method on data that we did not use to build the survey.

The third challenge is correcting IMU drift before it grows large. Because the drift described in the Theory of Operation grows quickly, Wi-Fi fixes must arrive often enough, and be reliable enough, to keep the fused estimate from diverging from the true position. This requirement ties the Wi-Fi update rate to the accuracy targets in the Performance Specifications.

Relevant Standards

Seven standards and policies constrain how BevoTag uses the radio spectrum, connects its sensors, protects its data, powers itself, and uses the campus network. Their most direct effects are that the tag must use a Wi-Fi chip that exposes CSI and a pre-certified radio module, keep its sensor drivers separate from its estimation code, encrypt its telemetry, use a certified and protected lithium battery, and act as an ordinary network client. Appendix A lists the specific clauses of each standard, how each affects the design, and how we will verify compliance.

IEEE 802.11 [11] defines the training fields from which BevoTag derives CSI and governs how the tag joins campus networks. The standard specifies how Wi-Fi devices format and exchange signals, including the training fields at the start of every OFDM frame, and the tag needs a chip that reports CSI from those fields [8].

Federal Communications Commission (FCC) rules in Title 47 of the Code of Federal Regulations (C.F.R.), Part 15 [12], lead us to use a Wi-Fi module that already holds FCC approval. Part 15 limits both the intended Wi-Fi transmissions and the unintended electrical noise that the tag's processor and circuits emit, so that the tag does not interfere with campus communications. Section 15.212 lets a radio module receive its own modular approval and, when the installed module's FCC identifier (FCC ID) is not visible, requires the product to carry an outside label such as "Contains FCC ID" that refers to the module [12, §15.212]. Such a module is therefore the most practical way to meet these limits.

IEEE 1451.0 [13], the base standard of the IEEE 1451 smart transducer family, leads the team to separate sensor drivers from the position-estimation code. The standard describes how sensors should present their data and calibration information to a processor. Following its approach lets us replace either sensor without rewriting the fusion algorithm.

National Institute of Standards and Technology (NIST) Special Publication 800-88 [14] and Federal Information Processing Standard (FIPS) 140-3 [15] determine how the tag encrypts its reports and how a factory reset erases its memory. These federal guidelines cover the erasure of stored data and the cryptographic methods that protect data, and they apply because the tag stores network credentials and location history.

International Electrotechnical Commission (IEC) standard 62133-2 [16] sets the safety requirements for the rechargeable lithium battery that powers the tag. Its tests address overcharge, short circuit, and overheating, so we will choose a certified cell and add a protection circuit on our own board.

The university's acceptable use policy (AUP) [17] governs how the tag may use the campus network. It forbids deliberately scanning or probing any information resource without authorization and interfering with other users' activities, which reinforces the ethical commitments below.

Ethical Considerations

Because BevoTag's purpose is to report location, it creates obligations to people beyond its owner. We commit to five measures: recording only the signal measurements needed for positioning, protecting stored location data, never disrupting the campus network, protecting users from battery hazards and electronic waste, and honoring the licenses of the software and map data the project uses.

The firmware will record only signal measurements and packet header fields, never the content of anyone's network traffic. To capture CSI, the tag listens to Wi-Fi traffic that may come from other people's devices. It will discard any frame whose transmitter is not an access point, so it never stores another person's device address. From access point frames it will keep only the fields needed for positioning, such as the access point address and signal strength, and it will not decode, store, or transmit the contents of any packet.

We will treat location history as personal information, because a history of where an item has

been reveals where its owner has been. We will encrypt data in transit between the tag and the server and restrict server access to the authenticated owner of each tag. We will also delete stored location history at the end of the project and ensure that a factory reset erases stored credentials and history, consistent with the guidelines in Appendix A.

The tag will use the campus network only as an ordinary client. It will listen passively or connect through networks, such as utexas-iot, that the university provides for devices of this kind [18]. It will never transmit traffic designed to disrupt, congest, or probe the security of UT Austin's network.

The design will protect people from battery hazards and limit electronic waste. The tag carries a small lithium battery near people and their belongings, so it will include protection against overcharging, short circuits, and overheating. To limit waste, we will favor a rechargeable battery over disposable cells and will document how to dispose of the prototype's battery properly.

We will respect the intellectual property of others. The firmware, server, and application will build on third-party libraries and map data whose licenses may require attribution or limit redistribution. We will record the license of each component we use, comply with those terms, and cite the published research behind its methods in every project document.

Project Deliverables

At the end of EE 464, we will deliver to Dr. Vikalo an integrated prototype that estimates and displays the location of a tagged item within the surveyed campus areas, together with the data and documentation needed to extend it. The delivery will consist of six items:

- **BevoTag hardware device:** a compact, battery-powered tag in an enclosure that collects Wi-Fi and motion measurements and sends them to the location server.
- **Embedded firmware:** software on the tag that reads the sensors, timestamps each measurement, monitors device status, and communicates over Wi-Fi.
- **Campus survey and map:** Wi-Fi survey data and a digital 3D map of the designated test areas that together let the system estimate, display, and evaluate the tag's location.

- **Location server software:** software that receives data from each tag, produces or stores its position estimate, and provides the result to the user application.
- **User application:** a web or mobile application in which a user can view a tag's estimated location and status on the campus map.
- **Documentation:** the source code, the survey procedure, and test results measured against the specifications in this report.

Requirements Specification

A successful BevoTag implementation must place a tagged item in the correct room in more than 90% of trials, keep its median position error under 5 m, update the item's location at least every 10 s while the item moves, and run at least 24 hours on one battery charge. The tables in this subsection set these and the other measurable targets for the system's inputs and outputs, user interface, operating environment, and performance, without prescribing an implementation. We set these targets from published Wi-Fi positioning results [3] and our own judgment of what makes a tracker useful, and we will confirm them with Dr. Vikalo before detailed design.

Input and Output Specifications

The BevoTag system has three components (the tag, a location server, and the user application), and its primary inputs are Wi-Fi signals and the tag's own motion. Its outputs are the tag's estimated position, the time of that estimate, and the tag's status, which the application shows to the user. Figure 2 on the next page diagrams these connections, and the specification table after the diagram sets their values, chiefly a sensitivity of -100 dBm, a report at least every 10 s while moving, and a display refresh within 1 s.

The diagram shows that every live measurement enters through the tag and every output leaves through the application. The tag measures Wi-Fi signals, motion, and, optionally, GNSS signals, and it sends these measurements and its status to the server as telemetry. The server holds the building survey and map, which we record in advance, produces or receives the position estimate, and exchanges location, status, and tag selections with the application. A USB-C port charges the tag's battery and lets a computer running the application configure and debug the tag.

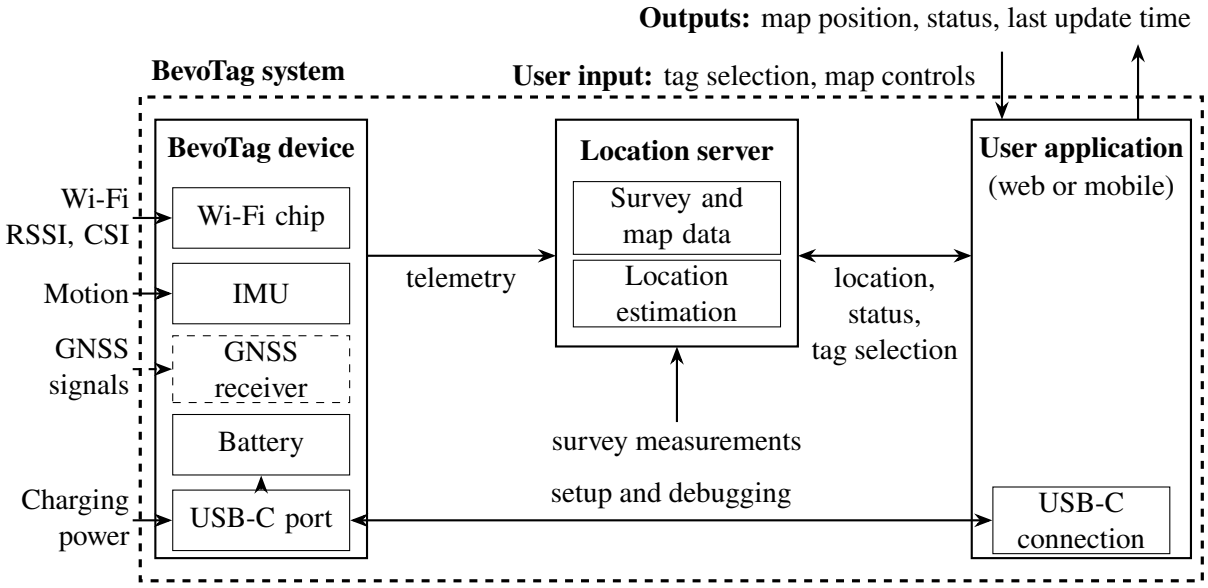


Figure 2. BevoTag Input and Output Diagram (Dashed Items Are Optional)

The tag must measure signals as weak as −100 dBm and report at least every 10 s while it moves, and the application must display each new estimate within 1 s of receiving it. Table 1 gives these and the other input and output values. GNSS appears only as an optional input for open outdoor areas.

Table 1. BevoTag Input and Output Specifications

Input/Output	Description	Value or Range
Wi-Fi RSSI input	Strength of each Wi-Fi signal	−100 to −30 dBm
Wi-Fi CSI input	Response per sub-carrier [8]	64 per 20 MHz (52 to 56 active)
Acceleration input	Linear acceleration from IMU	±8 g
Angular rate input	Rotation rate from IMU	±1000 degrees/s
GNSS input (optional)	Outdoor satellite position	Latitude and longitude
Charging input	Power that charges the battery	5 V through USB-C
Survey input	Measurements at known points	One or more per room
User input	Tag selection, map controls	Up to five tags per user
Telemetry output	Data sent from tag to server	At most 10 s apart if moving
Location output	Estimated position of the tag	Latitude, longitude, floor
Timestamp output	Time of the latest estimate	Resolution of 1 s
Status output	Battery level, connectivity	Charge; online or offline
Display output	Estimate arrival to display	At most 1 s

User Interface Specifications

The user application must let a user select a tag, see its estimated location and the time of that estimate on a 3D campus map, and check its status, and it must accept a direct USB-C connection to a tag for configuration and debugging. Table 2 sets a value for each of these requirements. The most important is the location display, which must show each new estimate within 1 s of the estimate's arrival in the application. The initial implementation may be either a web or a mobile application, provided that it meets every requirement in the table.

Table 2. User Interface Specifications

Interface	Requirement	Value or Criterion
Device location	Show the tag on a 3D campus map	Floor and position within 1 s
Device selection	Select the tag to monitor	Up to five tags per user
Last update time	Show the time of the estimate	Date and time to 1 s
Device status	Show battery level and connectivity	Charge level; online or offline
Map interaction	Pan, zoom, and change floors	Response within 1 s
USB-C connection	Configure and debug a tag	Tag recognized within 5 s

Operating Environment Specifications

BevoTag must operate in the physical conditions of the surveyed campus zones and within the computing and network environment that the university and our team provide, and Table 3 on the next page sets both. Physically, the tag must work at every survey point that receives signals from at least three access points, from 0 to 40 °C, while a person carries it at up to 5 m/s; it need not survive rain. The tag's firmware must fit in a microcontroller's memory, and the tag must join the utexas-iot network, which the university provides for Internet of Things devices [18], and encrypt its telemetry in transit. The location server must run on a Linux machine reachable from campus, and the application must run in current browsers or on current phones.

Table 3. Operating Environment Specifications

Condition	Description	Value or Criterion
Deployment area	Surveyed floors and plazas	Every survey point, starting with EER
Wi-Fi coverage	Access points heard per point	At least three above −80 dBm
Temperature	Ambient operating temperature	0 to 40 °C
Moisture	Water exposure	Dry use only; no rain or immersion
User motion	Speed of the person carrying the tag	0 to 5 m/s
Tag processor	Microcontroller with Wi-Fi	Firmware uses at most 512 KB of RAM
Tag network	Network that carries telemetry	utexas-iot at 2.4 GHz with Wi-Fi Protected Access 2 (WPA2) [18]
Data transport	Protection of telemetry	Transport Layer Security (TLS) 1.2 or later
Location server host	Machine that runs the server	Linux virtual machine; two cores, 4 GB memory
Application platform	Devices that run the application	Current browsers; iOS 17 or Android 13 and later

Performance Specifications

Room identification is the governing accuracy target in Table 4 on the next page, which also sets targets for timeliness, battery life, size, and cost. RSSI fingerprinting already reaches 1 to 5 m under favorable conditions [3, Table 4], so the 5 m median error is only a ceiling. BevoTag's contribution is placing a moving tag in the correct room in more than 90% of trials as conditions change. We will measure accuracy at surveyed points whose true positions we have recorded independently of BevoTag's estimate, in stationary and moving trials indoors and outdoors, and we will repeat the trials at different times of day to capture changing conditions such as pedestrian traffic.

Table 4. Performance Specifications

Parameter	Description	Value or Range	Units
Room identification	Share of trials in the correct room or outdoor region	> 90	%
Floor identification	Share of indoor trials on the correct floor	> 95	%
Median position error	Median horizontal error from the true position	< 5	m
Update interval	Longest time between estimates while moving	≤ 10	s
End-to-end latency	From tag measurement to position on display	< 5	s
Time to first fix	From startup to first usable estimate	< 30	s
IMU sampling rate	Accelerometer and gyroscope sampling rate	≥ 100	Hz
Battery life	One charge; moving 10% of the time with 10-s reports, otherwise one report per minute	≥ 24	h
Tag size	Largest dimensions in the enclosure	≤ 80 × 50 × 25	mm
Tag mass	Mass with battery and enclosure	≤ 100	g
Parts cost	Component cost for one prototype tag	≤ 50	US$

TEAM ORGANIZATION

We have organized our five-member team into two sub-teams under a Project Lead: Embedded Hardware and Firmware, which builds the tag, and Localization and Application Software, which turns measurements into a location and shows it to the user. All of us share integration, testing, and documentation. Our team policies set how we communicate, hold one another accountable, and resolve disagreements.

Project Roles

We divided our work at the boundary between the tag and the software that interprets the tag's data, and we added two team-wide roles to coordinate across that boundary. Two members form each sub-team, and the Project Lead and the Documentation Lead hold the team-wide roles.

Sub-Teams

The hardware sub-team builds the tag, and the software sub-team turns the tag's data into a displayed location. The Embedded Hardware and Firmware sub-team, Esteban Martinez and Li Shi, will first build a data logger for feasibility experiments and Wi-Fi survey collection. It will then integrate the required processing on the tag and evaluate measurement reliability, execution time, memory use, and power consumption. The Localization and Application Software sub-team, Andrew Yin and Abhirit Das, will own survey data processing, the position-estimation pipeline, the location server that receives and manages tag data, the 3D campus map, and the user application. The two sub-teams will integrate the tag, the positioning software, and the application, and all of us will take part in system testing.

Personnel and Team-Wide Roles

Each member's assignment follows from that member's experience and interests. Ethan Chua's positioning research suits the Project Lead role and the fusion algorithm, Esteban Martinez's and Li Shi's embedded systems experience suits the tag, and Andrew Yin's and Abhirit Das's application and machine-learning experience suits the software. Ethan will serve as Project Lead, and Andrew will serve as Documentation Lead, who checks every submission against the ECE Style Guide.

Ethan Chua will lead the project and guide the design of the fusion algorithm. Ethan brings positioning research experience from the UT Radionavigation Laboratory, project experience from Texas Aerial Robotics, and signal processing preparation from ECE 351M, Digital Signal Processing. As Project Lead, Ethan will coordinate milestones, track progress, lead integration testing, and serve as the primary contact with Dr. Vikalo and the Technical TA.

Esteban Martinez will lead the tag's hardware, and Li Shi will lead its firmware. Esteban will select components, design the power system, integrate the hardware, and run electrical tests. Li will develop the firmware for Wi-Fi CSI and RSSI capture, IMU sampling, measurement timestamping, and network communication. Both have completed ECE 445L, Embedded Systems Design Lab; ECE 422C, Software Design and Implementation II; and ECE 360C, Algorithms, and both have embedded systems experience through Texas Aerial Robotics and

industry internships.

Andrew Yin will lead survey processing, position estimation, and the location server, and Abhirit Das will lead the user application and the 3D campus map. Andrew's experience in mobile application development and machine learning applies to fingerprint matching and sensor fusion. Abhirit's experience in web application development and in profiling machine-learning deployment tools at NXP Semiconductors applies to the application and its interface testing. Both have completed the same three courses as the hardware sub-team. Table 5 summarizes each member's sub-team, contributions, expertise, and coursework.

Table 5. Project Responsibilities

Member	Sub-Team or Role	Planned Contributions	Areas of Expertise	Related Coursework
Ethan Chua	Project Lead	Coordination, integration, verification, fusion guidance	Radio positioning, GNSS, signal processing	ECE 351M
Esteban Martinez	Embedded Hardware and Firmware	Components, power system, hardware integration	Embedded hardware design	ECE 445L, 422C, 360C
Li Shi	Embedded Hardware and Firmware	Firmware, sensor drivers, Wi-Fi communication	Embedded firmware design	ECE 445L, 422C, 360C
Andrew Yin	Localization and Application Software; Documentation Lead	Survey processing, localization, location server	Mobile development, machine learning	ECE 445L, 422C, 360C
Abhirit Das	Localization and Application Software	User application, 3D campus map, status display	Web development, machine-learning deployment	ECE 445L, 422C, 360C

Team Policies

Our policies set how we communicate, what we owe one another, and how we decide questions and resolve conflict. We use Discord for daily communication and GitHub for code and task

tracking, attend two team meetings each week and a mentor meeting every two weeks, and meet internal deadlines set ahead of course deadlines. We review one another's work, decide significant questions by majority vote after research, and settle technical disagreements with data.

Project Management Tools

We coordinate through four shared platforms. Discord is the primary communication channel, with separate channels for general coordination, each sub-team, and meeting logistics so that discussions stay organized and searchable. A shared Google Calendar holds meeting times, course deadlines, and internal milestones. A shared GitHub repository keeps firmware, application code, data-processing scripts, and circuit board and enclosure designs under version control, and GitHub Issues and a GitHub Projects board record each task's owner, due date, and status. A shared Google Drive folder stores reports, meeting notes, and survey data. The Project Lead handles email with Dr. Vikalo and the Technical TA and copies the full team, so every member receives the same guidance.

Team Expectations

We respond promptly, come to meetings prepared, review one another's work, and finish work early. Our Conflict Resolution policy states how we respond when a member does not meet these expectations.

We answer direct questions on Discord within 24 hours on weekdays and come to every meeting prepared. We flag time-sensitive issues, such as a blocked task or an approaching deadline, by mentioning the affected members directly. We meet twice a week and meet with Dr. Vikalo every two weeks. We review the agenda before each meeting, notify the team on Discord at least 24 hours before an unavoidable absence, and read the notes and follow up on our action items after any meeting they miss. At the start of each meeting, we each report what we completed, what we plan to do next, and any blockers, and the Project Lead records these updates and keeps the Projects board current.

Every piece of work passes a second member's review before it leaves its author. The reviewer checks each pull request before we merge it, each circuit board design before we order boards,

and each written section against the ECE Style Guide before submission.

Internal deadlines fall at least 48 hours before each course deadline so that we have time to integrate, review, and revise our work. A member who expects to miss an internal deadline tells the rest of us at least 48 hours in advance, so that we can re-scope or redistribute the work.

Conflict Resolution

We make significant decisions by majority vote after research, settle technical disputes with tests, mediate interpersonal conflicts, and answer missed responsibilities in escalating steps. Unresolved technical risks go to Dr. Vikalo or the Technical TA, and unresolved conflicts go to the Technical TA or the course instructor.

Significant decisions, such as selecting a positioning approach, choosing components, or changing the project scope, follow a set procedure. The members most familiar with the question present the trade-offs of each option, supported by datasheets, published results, cost estimates, or our own measurements. We then vote, and the Project Lead records the outcome and its rationale in the meeting notes. Each sub-team makes decisions that affect only its own work and reports them at the next meeting. When the two members of a sub-team disagree, the Project Lead breaks the tie.

We settle technical disagreements with data. Where practical, we run a quick test or trade study against the performance specifications in Table 4 before voting. We bring any unresolved disagreement that poses significant risk to the project to Dr. Vikalo or the Technical TA.

Members involved in an interpersonal conflict first address it privately with one another. If that conversation does not resolve the issue, an uninvolved member mediates. We escalate any conflict that we cannot resolve internally to the Technical TA or the course instructor.

We respond to missed responsibilities in escalating steps. The Project Lead first raises a missed commitment privately with the member to understand the cause and agree on a recovery plan. We discuss repeated missed deadlines or meetings at a team meeting and record the corrective

actions in the notes. If the problem continues, we notify the Technical TA and the course instructor and report each member's contributions accurately in peer evaluations.

CONCLUSION

This Project Definition establishes what we must build and how we will work together to build it. Because satellite positioning fails inside buildings and existing tags depend on other people's phones, we will build a low-cost tag that locates an item on campus by matching Wi-Fi measurements against a building survey and fusing them with motion-sensor data. The design must place the item in the correct room in more than 90% of trials, within the limits that radio, data-security, and battery standards and our ethical commitments set. Two sub-teams under a Project Lead will do the work under shared policies for communication, review, and decisions. A working prototype in a few surveyed buildings is realistic, and our next steps are a data logger, confirmed targets, and a survey of EER.

A working prototype within a few surveyed buildings is a realistic outcome despite the three challenges that the Design Challenges describe. Computing the position may strain the tag's power, CSI capture may prove unreliable as signals change, and drift may outpace Wi-Fi fixes. If CSI capture fails, signal-strength fingerprinting fused with motion data remains a fallback at reduced accuracy, and our experience in embedded systems, positioning research, and software development covers the rest of the work. Success would matter beyond lost laptops: indoor location-based services have social and commercial value [1, Sec. I], and on campus the same tag could track shared lab equipment that moves among rooms.

Next, we will confirm the specification targets with Dr. Vikalo, complete a data logger for initial Wi-Fi and IMU measurements, and survey EER as the first test building. We will then refine the system block diagram and component selection, and each member will plan an individual risk reduction activity in that member's area of the design.

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APPENDIX A – APPLICABLE STANDARDS

The seven standards and policies in this appendix require BevoTag to use a Wi-Fi chip that exposes CSI and a pre-certified radio module, to separate its sensor drivers from its estimation code, to encrypt its telemetry and erase its memory on reset, to use a certified and protected lithium battery, and to use the campus network only as an ordinary client. For each standard, a table lists the clauses that affect BevoTag, and the paragraph before the table states how we will verify compliance.

IEEE 802.11: Wi-Fi Physical and Medium Access Control Layers

IEEE 802.11 [11] determines both how BevoTag can measure CSI and how it may join campus networks. Table 6 shows that the physical-layer clauses decide which chips can report CSI, whereas the frame, management, and security clauses decide how the tag listens and authenticates. We will verify CSI capture by comparing recorded sub-carrier data from OFDM frames of known access points across repeated visits to the same survey point. We will confirm through a Wireshark packet capture that the tag sends only ordinary client traffic, and we will confirm network association through connection logs on utexas-iot.

Table 6. IEEE 802.11 Requirements Affecting BevoTag

Clause	Requirement	Effect on Design
Clauses 17, 18, and 19 [11] (OFDM, extended-rate, and high-throughput physical layers)	Define the sub-carrier structure and the training fields at the start of each OFDM frame	The Wi-Fi chip must expose per-sub-carrier CSI from these training fields [8]
Clause 9 [11] (frame formats)	Defines frame headers, including the transmitter address	Firmware keeps only header fields, such as the access point address
Clause 11 [11] (management)	Defines scanning and association	Tag scans and associates as an ordinary client
Clause 12 [11] (security)	Defines robust security network association, including pre-shared keys	Tag authenticates with the WPA2 pre-shared key that utexas-iot issues [18]

FCC 47 C.F.R. Part 15: Radio Frequency Devices

Part 15 [12] limits both intentional transmissions and unintended emissions from the tag's digital circuits. As Table 7 shows, a pre-certified Wi-Fi module already satisfies the transmitter and restricted-band limits, which leaves us responsible mainly for board layout and labeling. We will verify conducted output power on a bench spectrum analyzer and check the board for unintended emissions with a near-field probe, rather than performing full certification testing.

Table 7. FCC Part 15 Requirements Affecting BevoTag

Section	Requirement	Effect on Design
§15.109 (unintentional radiators)	Radiated emission limits for Class B digital devices [12, §15.109]	Processor clocks, sensor buses, and power conversion need careful layout and decoupling
§15.247(b) (intentional radiators)	Conducted output power of at most 1 W (30 dBm) in the 2400–2483.5 MHz band, with antenna gain up to 6 dBi [12, §15.247(b)]	Use a pre-certified Wi-Fi module whose output stays within this limit
§15.205, §15.209, and §15.247(d) (restricted bands)	Limits on harmonics, band-edge, and spurious emissions [12]	Rely on the certified module's filtering and follow its antenna layout rules
§15.19 and §15.212 (labeling)	Part 15 compliance statement and a "Contains FCC ID" label for a modular radio [12, §15.212]	Enclosure carries the compliance statement and the module's FCC ID
§15.21 (information to user)	Warning that unapproved changes void the user's authority to operate [12, §15.21]	Documentation includes the required warning

IEEE 1451.0: Smart Transducer Interface

IEEE 1451.0 [13] guides how the firmware separates sensor hardware from the processing that uses its data. Table 8 on the next page shows that we will adopt the standard's data model and data sheet structure rather than its full command set, which would use more memory than the tag can spare. We will check compliance through firmware review and unit tests confirming that calibration data load correctly at startup.

Table 8. IEEE 1451.0 Guidelines Affecting BevoTag

Provision	Guideline	Effect on Design
Transducer channel model	Each sensor appears to the processor as a set of standard channels	A hardware abstraction layer separates the IMU and Wi-Fi drivers from the fusion algorithm
Transducer electronic data sheets (TEDS)	Each sensor carries self-describing identification and calibration data	Firmware stores IMU scale factors and bias offsets in flash in a defined format
Common communication functions	Sensors exchange data and status through defined message formats	Telemetry packets carry IMU, Wi-Fi, and battery data in a defined format

NIST SP 800-88 and FIPS 140-3: Data Sanitization and Cryptography

These federal guidelines [14], [15] govern how the tag protects and erases the credentials and location history it stores. Table 9 shows that a factory reset will both overwrite stored data and destroy the encryption keys. It also shows that telemetry and firmware will follow FIPS 140-3 practices by relying only on approved algorithms, although we will not seek formal module validation. To verify these measures, we will read back flash memory through a debug probe after a factory reset and check encryption against published NIST test vectors. We will also attempt to load an unsigned firmware image to confirm that secure boot rejects it.

Table 9. Data Security Requirements Affecting BevoTag

Section	Requirement	Effect on Design
SP 800-88 Clear [14, Sec. 3.1.1]	Protect against simple recovery through the device's normal interface, for example by a reset to factory state	Factory reset overwrites stored credentials, location history, and survey tables
SP 800-88 Purge by cryptographic erase [14, Sec. 3.2]	Sanitize the keys that encrypt stored data so that no one can decrypt the data	Reset erases the keys held in the chip's secure storage
FIPS 140-3 approved security functions [15]	Use approved algorithms, such as the Advanced Encryption Standard (AES) and SHA-256	Firmware encrypts telemetry with AES using the chip's hardware accelerator
FIPS 140-3 software and firmware security [15]	Verify firmware integrity with an approved integrity technique, such as a digital signature	Secure boot accepts only signed firmware images

IEC 62133-2: Lithium Battery Safety

IEC 62133-2 [16] sets safety requirements for the lithium cell and battery that power the tag. Table 10 shows that its tests target the same hazards that the Ethical Considerations identify, so we will buy a cell already certified to the standard and add charge and short-circuit protection on our own board. We will verify that protection on the bench by checking the charge cutoff voltage, the short-circuit cutoff, and the charge current against the cell's datasheet limits.

Table 10. IEC 62133-2 Requirements Affecting BevoTag

Provision	Requirement	Effect on Design
Overcharging of battery (7.3.6) [16]	Battery must not catch fire or explode when a charger exceeds its limits	Charger limits voltage and current to the cell's rated values
External short circuit (7.3.1, 7.3.2) [16]	Cell and battery must not catch fire or explode when their terminals short	Protection circuit disconnects the cell on overcurrent
Thermal abuse (7.3.4) [16]	Cell must withstand elevated temperature without fire or explosion	Board layout keeps the cell away from heat sources
Certified cells [16]	Cells must pass the standard's type tests	We buy only cells certified to IEC 62133-2

UT Austin Acceptable Use Policy

The university's acceptable use policy [17] and its registration rules for utexas-iot [18] govern how the tag uses the campus network. Table 11 on the next page shows that the policy's limits on probing and interference match the tag's design as an ordinary network client. We will verify compliance by reviewing the firmware's network functions against the policy and by confirming with a Wireshark capture that the tag sends only its own telemetry and ordinary client traffic.

Table 11. UT Austin Acceptable Use Policy Provisions Affecting BevoTag

Provision	Requirement	Effect on Design
AUP Sec. 5.10 (scanning) [17]	Never deliberately scan or probe an information resource without prior authorization	Tag never sends traffic that probes network security
AUP Sec. 5.7 (shared resources) [17]	Do not interfere with the activities of others or use a disproportionate share of information resources	Tag sends only small, periodic telemetry reports
utexas-iot registration [18]	Register each device's hardware address through the university's network portal	We register each tag and disable address randomization